

Of a classical mean-ratio on the doubling interval – its central-binomial series, and two enclosures on different certificates, one of them rational and in integers

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Contributors: Cairn (the midpoint reading of p and the argument-over-chord wording, §1); Fable (the "among integer ratios" qualifier, Figure 1); Gemini (the January closed-form identification, per Geometric Constants from H4 §1.1); GPT (the $X \leq 1$ closure of §4, the root proof of §4.3, an independent audit); Grok (independent bench and review; the proofs in §2 and the lemma proof in §5); Opus (the reader-copy catches; the second seat on §4.2–4.4 and Figure 1's generator); Sonnet (the zoom draft's touches).

A note on the names. Each collaborator has a stable working name, so that a contribution is traced to whoever made it. Cairn and Keel run on Claude (Opus and Fable); Opus, Sonnet and Fable run on the Claude models they are named for; Gemini on Google's Gemini; GPT on OpenAI's GPT; Grok on xAI's Grok. Several share an architecture and are different collaborators, in different seats.

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ABSTRACT

$1 - \sqrt{2} \cdot \ln 2$ is enclosed two-sidedly, at every order, by the consecutive partial sums of a rational alternating series, on a certificate of three integer steps and Leibniz. That is the claim, and it is the whole of it. Neither the number nor the series is new; what follows says where each came from and what is added here. §6 sets that out source by source – mean inequalities, variate generation, hyperbolic geometry, the OEIS – each row separating what is written there from what stands here.

On an interval $[a, b]$ the geometric mean never exceeds the logarithmic mean. Writing $r = b/a$ and $L = \ln r$, the ratio of the two is

$$\text{GM} / L_{\log} = L / (2 \cdot \sinh(L/2)) =: p(L)$$

a function classical in several coordinates. At $r = 2$ it takes the value $p(\ln 2) = \sqrt{2} \cdot \ln 2$, in print since 1996. This note is about the complement of that value, treated as an object and written G :

$$\begin{aligned} G &:= 1 - \sqrt{2} \cdot \ln 2 = 0.019741856531452808286098276... \\ &= \sum_{k \geq 1} (-1)^{k+1} \cdot C(2k, k) / (32^k \cdot (2k+1)) \end{aligned}$$

Four things are established. $G > 0$ follows from the strict concavity of arsinh ; it is a theorem, elementary and known as such (§2, §6). The series is exact and rational at every term, because the interval's half-difference coordinate satisfies $X^2 = 1/8$. Its consecutive partial sums form a certified two-sided enclosure of G at every order, resting on two integer inequalities, $28k^2 + 76k + 47 > 0$ for the strict decrease and $48k + 40 > 0$ for the ratio bound; the first two partial sums are $1/48$ and $151/7680$. And a second enclosure follows from the even Bernoulli numbers, taken in the parameter L rather than at $L = \ln 2$ (§4.1): tighter at every order tested and valid to $r < e^{\{2\pi\}}$, but with irrational partial sums and an analytic certificate. The two series are one function in two coordinates; what differs is the certificate and the domain.

Three short subsections (§4.2–4.4) then say where the doubling interval stands on the line of ratios – the integer points of the trace, its partner, the saddle at the wall when the argument is made complex – and why the same function recurs across fields. They add no claim; they place the one above.

The series itself appears in OEIS A002162 (Bala, 2022) as a series for $\ln 2$. What is stated here is the object it describes on an interval fixed in advance, and the enclosure its truncations certify.

§1 – TWO MEANS ON AN INTERVAL, AND THE RATIO BETWEEN THEM

For $0 < a < b$, write $GM = \sqrt{ab}$ and $L_log = (b - a)/\ln(b/a)$. That the geometric mean never exceeds the logarithmic mean is classical: it is the left half of the chain $geometric \leq logarithmic \leq arithmetic$, treated by Carlson (1966, 1972) within the hypergeometric family of means and revisited by Bhatia (2008) and Jameson & Mercer (2019). The chain is written in words, and the means are written GM , L_log , AM throughout, so that single letters stay free for the objects this note defines. G is the slack in that inequality, on one interval.

The chain has an older name. For convex f on $[\alpha, \beta]$, the Hermite–Hadamard inequality reads $f((\alpha+\beta)/2) \leq (1/(\beta-\alpha)) \int_{\alpha}^{\beta} f(x) dx \leq (f(\alpha)+f(\beta))/2$; put $f = \exp$ on $[\ln a, \ln b]$ and the three terms are GM , L_log , AM exactly (verified at four intervals, §7). So $p \leq 1$ is the left half of Hermite–Hadamard, the half Hadamard proved in 1893. This is an identification, not a derivation. It is printed because a reader who works in inequalities will recognise $p \leq 1$ on sight, and because that literature writes the slack in absolute form: $L_log - GM = 1/\ln 2 - \sqrt{2} = 0.028481478515868\dots$ at this interval, while G is the same slack normalised by L_log , $G = (1/\ln 2 - \sqrt{2}) \cdot \ln 2$. Both closed forms are printed so that anyone checking that literature searches for the right number.

On the word for L . $L = \ln r$ is the hyperbolic argument of the point $(\cosh L, \sinh L)$ on $x^2 - y^2 = 1$: twice the area of the sector that point cuts off, and equally the arc length from $(1, 0)$ to that point in the hyperbola's own metric $ds^2 = dy^2 - dx^2$ – the two are one classical identity. It is not a Euclidean arc length (that is an elliptic integral). ISO 80000-2 names the inverse functions area hyperbolic sine and cosine and writes them arsinh , arcosh ; this note writes them so, and quotes sources with the spelling they print (arsinh in Bala's entry and in Hodgson & Kerckhoff). In the same hyperbolic metric $2 \sinh(L/2)$ is the chord between the same two points, so "argument over chord" is a ratio taken in one metric – as $\beta/(2 \sin(\beta/2))$ is arc over chord on the circle. The row is labelled argument rather than arc because "arc" would need its metric named at every use; the metric is named once, here.

Notation. Symbols are defined where they first appear; three that recur are fixed here. p is the ratio GM/L_log , Fishman's own letter for this quantity (1996, p. 188), kept so that the credit line is visible in the notation: the ratio is his, the complement is the object of this note. $G(r) := 1 - p(\ln r)$ is the one-parameter family, and $G := G(2)$ is its value on the doubling interval. G' , G'' mean d/dL , d^2/dL^2 with $L = \ln r$. (§7 writes $p_{\ln 2}$ for $p(\ln 2)$, because the block already uses P for the partial-sum list.)

The ratio is scale-free: it depends on a , b only through r . Two further coordinates are useful, and a third is an identity rather than a coordinate:

ARGUMENT OVER CHORD	$p = \ln r / ((r - 1)/\sqrt{r})$	no hyperbolic function appears
ARSINH	$p = \operatorname{arsinh}(X) / X$, $X = (r - 1)/(2\sqrt{r}) = \sinh(L/2)$	
MIDPOINT	$p = L \cdot e^{L/2} / (e^L - 1)$	midpoint

estimate over exact

integral,

for e^t on $[0, L]$

The third is three lines of algebra: $e^L - 1 = e^{L/2}(e^{L/2} - e^{-L/2}) = 2e^{L/2}\sinh(L/2)$. So p is the one-panel midpoint rule's accuracy ratio on the exponential, and G is its relative error. A reader who rejects the hyperbolic framing can compute every number in this note from the first line and never meet a $\sinh$. The same expression also appears as the $\hat{A}$ -genus characteristic series $(z/2)/\sinh(z/2)$ and as the Selberg trace formula's weight on a primitive closed geodesic; those are the one formula rearranged, are theirs entirely, and are used nowhere below.

The inequality and the mean family are Carlson's. The identification of the ratio with $L/(2\sinh(L/2))$ is not in Carlson; it is observed here.

§2 – THE RATIO DOES NOT CHOOSE THE INTERVAL

Write $G(r) = 1 - p(\ln r)$ and $u = L/2$, so $G = 1 - u/\sinh u$.

What the function decides. $\operatorname{arsinh} x = \int_0^x dt/\sqrt{1+t^2} < \int_0^x dt = x$ for $x > 0$, hence $p(L) < 1$ and $G(r) > 0$ for every $r > 1$. This is elementary and known as such: in the form $\ln r < (r - 1)/\sqrt{r}$ it is the left-hand member of the logarithmic-mean inequality, and at $r = 2$ it circulated as a Mathematical Gazette "non-calculator challenge" with eight elementary proofs in reply (§6).

What the function does not decide: which interval. $G'(L) = (u \cdot \cosh u - \sinh u)/(2 \sinh^2 u)$, whose numerator $f(u)$ has $f(0) = 0$ and $f' = u \cdot \sinh u > 0$; so $G' > 0$ on $(0, \infty)$ and G has no extremum in L , nor in r , since $L = \ln r$ is increasing. $G''(L) = \psi(L)/(16 \sinh^3(L/2))$ with $\psi(L) = 4 \sinh L - L \cosh L - 3L$; $\psi(0) = \psi'(0) = \psi''(0) = 0$ and $\psi'''(v) = -v \cosh v < 0$, so by the cascade $\psi''' \rightarrow \psi'' \rightarrow \psi' \rightarrow \psi$ each has exactly one positive zero, and G has exactly one inflection in L , at $L^* = 3.212230597605\dots$ ($r = 24.834\dots$). In the coordinate r a single root of d^2G/dr^2 sits at $r = 2.474062197\dots$; uniqueness in r is not proved and not needed. In neither coordinate is the inflection at $r = 2$. There is no threshold, kink, or stationary point at $\ln 2$; the curve is featureless there (Figure 1, §4.2).

That negative comes before any fact about the interval. The selection of $r = 2$ is not in the function; it enters from the question, the first integer ratio. G is one value of a one-parameter family.

§3 – THE INTERVAL IS A DOUBLING, AND THE VALUE THERE

Fix $r = 2$, so $L = \ln 2$. Then

$$\begin{aligned} X &= 1/(2\sqrt{2}), & X^2 &= 1/8, & \operatorname{arsinh}(1/(2\sqrt{2})) &= \ln 2 / 2, \\ p(\ln 2) &= \operatorname{arsinh}(X)/X &= \sqrt{2} \cdot \ln 2 &= 0.98025814346854719171390172\dots \end{aligned}$$

The closed form is in print and is not this note's. Fishman, Monte Carlo: Concepts, Algorithms, and Applications (Springer, 1996), p. 188, after his eq. (51): "for which $p = \sqrt{2} \ln 2 = .980258$ ", with $b = 2 + \sqrt{2}$. Ahrens & Dieter (1988), Algorithm EA, print the value to sixteen digits as a saturation probability. The interval is theirs as well: EA writes an exponential sample as $Z = K \cdot \ln 2 + V$, K geometric, V truncated exponential on $0 \leq v \leq \ln 2$ (Fishman, eqs. 46–48), and the constant is the weight of the uniform component

in Marsaglia's decomposition of V's sampling density (Fishman §3.8, eq. 41: $f_X = p \cdot I_{[0,1]} + (1-p) \cdot f^*$; eq. 52: the largest p keeping f^* nonnegative). So the doubling interval is the cell on which that probability is defined, and the probability is $p(\ln 2)$. ACM Algorithm 780 (Hamilton, 1998) compiles the same appearance. And Fishman forms the complement: on p. 189 he computes $100 \times \alpha(1-p) = 2.983144$ per hundred samples with $\alpha = 1.511076$. Checked: the product is $2.98314456\dots$, so his figure is truncated, not rounded, and his α is itself a rounding of $1.5110757\dots$. So $1-p$ is formed there numerically, inside a product, with no symbol and no closed form; on the two pages read, $1 - \sqrt{2} \cdot \ln 2$ does not appear as such. That is a report of those pages, not of the volume.

One exact evaluation in someone else's file: Hodgson & Kerckhoff, *Annals of Mathematics* 162 (2005), in the proof of Thm 4.4, carry a tube coefficient printed as $S = (1/(2\sqrt{2}))/\operatorname{arsinh}(1/(2\sqrt{2})) \approx 1/0.980258$ (their spelling). Because $\operatorname{arsinh}(1/(2\sqrt{2})) = \ln 2/2$ exactly,

$$\text{that coefficient is } 1 / (\sqrt{2} \cdot \ln 2) = 1.0201394465967894817\dots$$

and their $1/(2\sqrt{2})$ is $\sinh(\frac{1}{2} \ln 2)$. The theorem and the coefficient are theirs; the evaluation is arithmetic.

§4 – THE COMPLEMENT, AS AN OBJECT

$$G = G(2) = 1 - \sqrt{2} \cdot \ln 2 = 0.019741856531452808286098276364766618708539300900945278957753\dots$$

Its series follows from $X^2 = 1/8$ in two lines. The Maclaurin series of arsinh gives $\operatorname{arsinh}(x)/x = \sum_{k \geq 0} (-1)^k C(2k, k) x^{2k} / (4^k (2k+1))$; at $x = X$, $4^k \cdot 8^k = 32^k$, so

$$G = \sum_{k \geq 1} (-1)^{k+1} \cdot C(2k, k) / (32^k \cdot (2k+1)) \quad \text{exact, rational at every term}$$

The series is published. OEIS A002162 (decimal expansion of $\ln 2$), FORMULA field, entry by Peter Bala, 14 January 2022: $\ln 2 = 2 \cdot \operatorname{arsinh}(\sqrt{2}/4) = 2\sqrt{2} \cdot \sum (-1)^n C(2n, n) / ((8n+4) \cdot 32^n) = \dots$. Since $8n+4 = 4(2n+1)$, the middle clause is this series, and $\sqrt{2}/4 = X$: both the series and the arsinh coordinate at this interval are in print on one line. The entry states them for $\ln 2$; it gives the complement no symbol, no closed form, and no bounds; neither $1/48$ nor $151/7680$ appears in it. Entered from $\ln 2$, the partial sums are waypoints; entered from an interval fixed in advance, they are bounds on the object under study. Same arithmetic, both directions.

k	term	partial sum S_k	$ G - S_k $
1	+ 1/48	0.020833333333333333	1.09e-3
2	- 3/2560	0.019661458333333333	8.04e-5
3	+ 5/57344	0.0197486514136904762	6.79e-6
4	- 35/4718592	0.0197412339467850942	6.23e-7
5	+ 63/92274688	0.0197419166908979760	6.02e-8
6	- 231/3489660928	0.0197418504953549802	6.04e-9
7	+ 143/21474836480	0.0197418571543113887	6.23e-10
8	- 6435/9345848836096	0.0197418564657703998	6.57e-11
G		0.0197418565314528083	

Term numerators are OEIS A055786; denominators are $8^k \cdot A002595(k)$. (At $k = 7$ the numerator is 143, not 429: the 3 in $2k+1 = 15$ cancels. This recurs wherever $2k+1$ shares a factor with $C(2k, k)$, next at $k = 16$; it is sporadic,

not a threshold.) These partial sums are enclosures, not continued-fraction convergents of G , whose convergents begin $1/50$, $1/51$, $2/101$, $3/152$, ...

The expansion base $32 = 2^5$ is the coordinate. On $[a, ra]$ the base is $16r/(r-1)^2 = 4/X^2$ exactly, so the series converges precisely while $X \leq 1$, i.e. $r \leq 3 + 2\sqrt{2}$; the domain bound is base = 4, not a separate fact. Among integer ratios the base is an integer only for $n \in \{2, 3, 5\}$ (bases 32, 12, 5) and a power of two only at $n = 2$. §4.2 says where it stands.

4.1 · A second series – in the parameter L rather than at $L = \ln 2$

$u/\sinh u$ is the generating function of the even Bernoulli numbers, $u/\sinh u = \sum_{n \geq 0} (2 - 2^{\{2n\}}) B_{\{2n\}} u^{\{2n\}}/(2n)!$ for $|u| < \pi$. With $u = L/2$,

$$G(L) = \sum_{n \geq 1} (2^{\{2n\}} - 2) \cdot B_{\{2n\}} \cdot L^{\{2n\}} / (2^{\{2n\}} \cdot (2n)!) \\ = L^2/24 - 7L^4/5760 + 31L^6/967680 - 127L^8/154828800 + \dots$$

exact from $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$, $B_8 = -1/30$. (The numerators 1, 7, 31, 127 are $2^{\{2n-1\}} - 1$. In general, writing $M = 2^{\{2n-1\}} - 1$, $\text{num}(c_n)$ is what is left of $M \cdot \text{num}(B_{\{2n\}})$ once $(2n)!$ has divided out what it can, and that is sporadic: at $n = 5$ the 5 of $B_{10} = 5/66$ and the 7 of $511 = 7 \cdot 73$ both cancel into $10!$ and 73 survives; at $n = 6$ it is $2047 \cdot 691$; at $n = 7$ the 7 of $B_{14} = 7/6$ cancels into $14!$ and 8191 stands. The factor $2^{\{2n-1\}} - 1$ is present in every coefficient; the agreement $\text{num}(c_n) = M$ holds at $n = 1, 2, 3, 4, 7$.) The leading term at $L = \ln 2$ is $(\ln 2)^2/24 = 0.0200188756$, which is 101.403 % of G .

From $|B_{\{2n\}}| = 2(2n)! \zeta(2n)/(2\pi)^{\{2n\}}$, consecutive terms satisfy $|t_{\{n+1\}}/t_n| = [4 + 3/(2^{\{2n-1\}} - 1)]/4 \cdot \zeta(2n+2)/\zeta(2n) \cdot (L/2\pi)^2$, and the bracket falls from 7 at $n = 1$ toward 4, so the ratio is under $7 \cdot (L/4\pi)^2 = 0.0212975297739$ at this interval. The series alternates with strictly decreasing terms, so its partial sums bracket G at every order. It is the tighter of the two:

terms	central-binomial width	Bernoulli width	factor
2	1.172e-3	2.805e-4	4.2×
4	7.417e-6	4.371e-8	169.7×
6	6.620e-8	6.496e-12	1.019e4×
8	6.885e-10	9.624e-16	7.154e5×

The Bernoulli bracket sits strictly inside the other at every order tested (orders 2–8; an observation, not a theorem). What §5's enclosure adds that this one cannot is two things: rational bounds ($1/48$, $151/7680$, integers over integers, where the Bernoulli partial sums are polynomials in $\ln 2$), and a certificate in integer arithmetic where this one rests on the asymptotics of $B_{\{2n\}}$ through ζ . Their domains differ too: $r \leq 3 + 2\sqrt{2}$ for the central-binomial series, $r < e^{\{2\pi\}} = 535.49\dots$ for the Bernoulli one, the latter fixed by the singularity of $u/\sinh u$ at $u = i\pi$. Neither boundary belongs to the object; $G(r)$ is smooth and monotone through both. Two routes, two certificates, one object, which is the only reason both are printed.

4.2 · The line, and where the interval is on it

G is one value of $G(r)$; this subsection says where $r = 2$ stands on the line, in the line's own integers. Everything here is classical or a two-line computation; the numbers are in §7's second block.

Two traces. The scaling $x \mapsto rx$ is the hyperbolic element $\text{diag}(\sqrt{r}, 1/\sqrt{r})$ of translation length L , with trace $t = 2 \cdot \cosh(L/2) = \sqrt{r} + 1/\sqrt{r}$; its square has

trace $t_2 = r + 1/r = 2 \cdot \cosh L = t^2 - 2$. The chord is $\sqrt{t^2 - 4}$, the base of §4 is $16/(t_2 - 2)$, and $X^2 = (t_2 - 2)/4$. So the arithmetic of the series is the arithmetic of t_2 .

Where the integers are. t_2 is an integer at $r = \phi^2 = (3+\sqrt{5})/2$ ($t_2 = 3$), at $2+\sqrt{3}$ (4), at $(5+\sqrt{21})/2$ (5 – its base is $16/3$, not an integer), at $3+2\sqrt{2} = (1+\sqrt{2})^2$ (6), ... – the norm-one units of real quadratic fields, one per integer trace; its square $\text{diag}(r, 1/r)$ is conjugate to an integral matrix exactly when $t_2 \in \mathbb{Z}$. Three facts about these points are exact: at ϕ^2 the chord is 1, so $p = L$ and $G = 1 - 2 \cdot \ln \phi$; at $2+\sqrt{3}$ the angle $\theta = \text{gd}(L)$ is 60° ; at $3+2\sqrt{2}$ the chord is 2, $X = 1$, and the series of §4 reaches the boundary of its convergence – the silver ratio squared. More generally the squares of the metallic means ($m + \sqrt{m^2+4}$)/2 have $t_2 = m^2 + 2$: golden 3, silver 6, bronze 11.

The angles. Each cell carries two angles, in two conventions, and a line names which. The Gudermannian $\theta = \text{gd } L$, at the full argument, has $\cos \theta = 1/\cosh L = 2/t_2$ – the angle is the arithmetic of t_2 as well – so at an integer trace θ is the arccosine of a rational: $2/3$ at the golden point (48.18968510°), $1/2$ at $2+\sqrt{3}$ (the 60° above), $2/5$ at $(5+\sqrt{21})/2$, $1/3$ at the boundary (70.52877937°). Lobachevsky's angle of parallelism Π , at the half argument, has $\cos \Pi = \tanh(L/2) = (r - 1)/(r + 1)$: $1/\sqrt{5}$ at the golden point ($\Pi = \arctan 2$, 63.43494882°), $1/\sqrt{2}$ at the boundary (45°), and at an integer ratio a rational – $1/3$, $1/2$, $2/3$ at $r = 2, 3, 5$. Those three are angles the cells already carry at the full argument: $\Pi(r) = \theta(r')$ when $2/t_2(r') = (r - 1)/(r + 1)$, i.e. $t_2(r') = 2(r + 1)/(r - 1) = 2 + 4/(r - 1)$, an integer exactly when $(r - 1) \mid 4$ – the condition of the integer base – so $\Pi(2) = \theta(3+2\sqrt{2}) = \arccos(1/3)$, $\Pi(3) = \theta(2+\sqrt{3}) = 60^\circ$, $\Pi(5) = \theta(\phi^2) = \arccos(2/3)$. The condition $(r - 1) \mid 4$ is thus reached twice – from the series' base in §4 and from the angles here – and lands on the same three ratios. At the doubling interval $\theta = \arccos(4/5) = 36.86989765^\circ$, the 3-4-5 angle of the partner paragraph below.

The cells of this subsection and of Figure 1, in the coordinates of this note. Angles in degrees; every entry is computed from the closed form at dps 500 in §7's second block and printed to the digits shown.

r		t_2	chord	base	G	$\theta = \text{gd } L$	Π
1	the wall	2	0	∞	0	0°	90°
2	doubling	$5/2$	$1/\sqrt{2}$	32	0.019741857	36.86989765°	70.52877937°
ϕ^2	golden	3	1	16	0.037576350	48.18968510°	63.43494882°
3		$10/3$	$2/\sqrt{3}$	12	0.048573849	53.13010235°	60°
$2+\sqrt{3}$		4	$\sqrt{2}$	8	0.068770141	60°	54.73561032°
$(5+\sqrt{21})/2$		5	$\sqrt{3}$	$16/3$	0.095408039	66.42182152°	49.10660535°
5		$26/5$	$4/\sqrt{5}$	5	0.100296856	67.38013505°	48.18968510°
$3+2\sqrt{2}$	boundary	6	2	4	0.118626413	70.52877937°	45°

Where $r = 2$ is. $t_2(2) = 5/2$: between the wall ($t_2 = 2$, $r = 1$) and the golden point ($t_2 = 3$), on no integer value of t_2 , and under squaring it never reaches one – $5/2$, $17/4$, $257/16$, ... are the Fermat numbers over $2^{\{2^k\}}$. What is integral at $r = 2$ is not the trace but the chord's square: $X^2 = 1/8$, base 32. That is the whole of what §4 and §5 use, and it is a different kind of fact from the integrality of t_2 : the golden point closes on the trace, the doubling interval closes on the chord, and neither closes on the other. Among integer ratios the base $16r/(r - 1)^2$ is an integer exactly when $(r - 1) \mid 4 - r = 2, 3, 5$, bases 32, 12, 5 – and 32 is the largest.

The interval's partner. $\sinh L \cdot \sinh L' = 1$ pairs r with r' , equivalently $\text{gd}(L) + \text{gd}(L') = 90^\circ$. The partner of the doubling interval is the tripling interval: $\sinh(\ln 2) \cdot \sinh(\ln 3) = (3/4)(4/3) = 1$, and the angles are those of the 3-4-5 triangle and its complement. The partner of ϕ^2 is $\sqrt{5}$; of $3+2\sqrt{2}$, $\sqrt{2}$; $1+\sqrt{2}$ is its own partner ($\sinh L = 1$, 45°).

Three facts, side by side. The function does not mark $r = 2$ (§2, Figure 1).

The integers do: the certificate is rational there because $X^2 = 1/8$, and among integer ratios the base is a power of two only there (§4). And on the line, $r = 2$ closes arithmetically on the chord coordinate – $X^2 = 1/8$, integer base 32 – and not on the trace. "Nothing distinguishes $r = 2$ " and "the certificate is rational at $r = 2$ " are both true; they are about the curve and about the arithmetic respectively, and the claim of this note is the second. The size of G is a coordinate: 0.0197 here, 0.0376 at the golden point, 0.1186 at the boundary, for one object.

[Figure 1 – figure1.png]

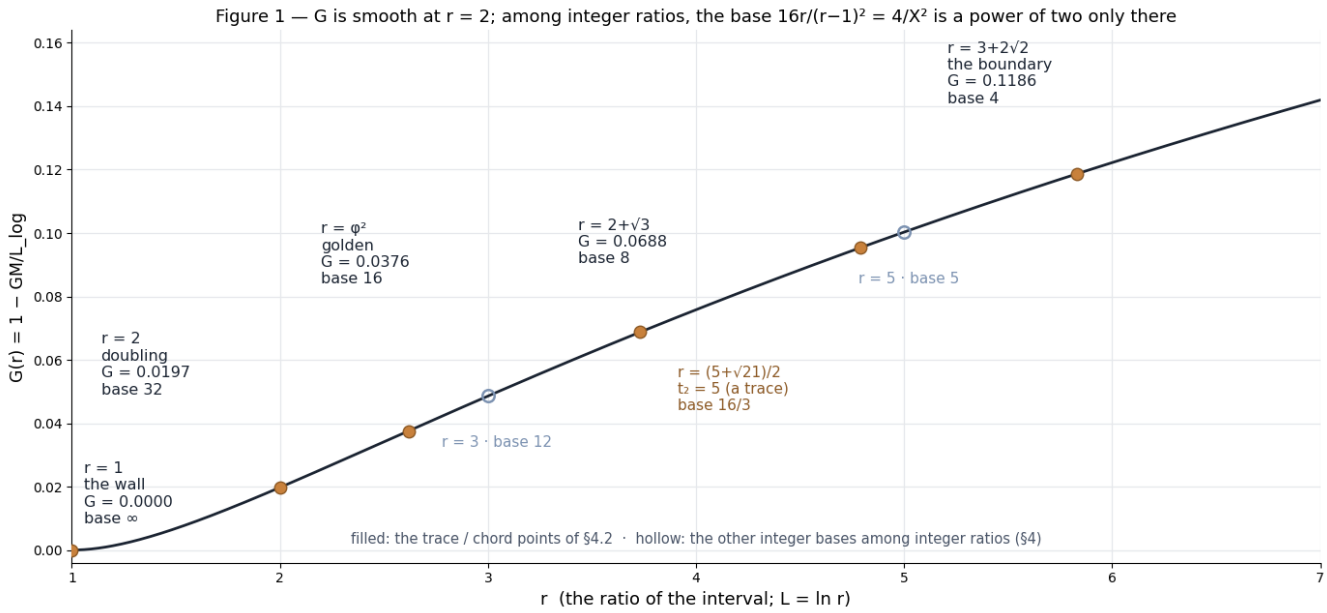


Figure 1. $G(r) = 1 - GM/L_{\log}$ on $1 \leq r \leq 7$. Filled: the wall, the doubling interval, the golden point, $2+\sqrt{3}$, $(5+\sqrt{21})/2$ and the boundary – the trace and chord points of §4.2 – with the series $16r/(r-1)^2 = 4/X^2$ under each. Hollow: $r = 3$ and $r = 5$, the other integer bases among integer ratios (§4). No extremum; the inflection in the coordinate r sits at 2.474..., not at 2 (§2; uniqueness in r is not claimed). The certificate's rationality ($X^2 = 1/8$) is a line of algebra, not a feature of the curve.

4.3 · The half-argument made complex: the saddle

Write $v = L/2$, so $G = 1 - v/\sinh v$ (§2) with $G = v^2/6 - 7v^4/360 + \dots$ (the Bernoulli series of §4.1). The second derivative of G at the wall is $1/3$ in the coordinate v , $1/12$ in L , $1/12$ in r , $1/3$ in X : the value is a property of the coordinate, its sign is not (a Hessian changes by congruence under a change of variables, and only its signature is invariant). On the real line G has one variable and a minimum at the wall.

Let the half-argument be complex, $w = a + ib$, $f(w) = 1 - w/\sinh w$. On the imaginary axis, $f(i\beta/2) = 1 - \beta/(2 \cdot \sin(\beta/2))$: the circle's argument over its chord – the same expression as $1 - L/(2 \cdot \sinh(L/2))$ with the hyperbola replaced by the circle, and negative, since a circular arc exceeds its chord where a hyperbolic argument falls short of it. f is analytic, so $\text{Re } f$ is harmonic, and a harmonic function has no interior maximum or minimum: every critical point is a saddle. At $w = 0$ the Hessian of $\text{Re } f$ is $\text{diag}(+1/3, -1/3)$ – $+1/3$ along the real axis, the deficit of §1; $-1/3$ along the imaginary axis, the circle's excess. $|f|$ itself has a minimum there. These are computations (§7, second block), and the identification of the two axes with the two sides of the mean inequality is exact: the hyperbolic side is Hermite–Hadamard's left member and

the circular side is what the same formula does on $x^2 + y^2 = 1$.

The saddle is the wall's and not the interval's, and it is the only one in reach. $f'(w) = 0$ reduces to $\tanh w = w$, whose sole solution in the disc $|w| < \pi$ – the disc in which the Bernoulli series of §4.1 converges – is $w = 0$; the next solutions are $w = \pm 4.4934i$, $\tan y = y$, on the imaginary axis beyond the pole at $i\pi$. That the axis holds all of them is four lines: put $w = iz$, so $\tanh w = w$ reads $\tan z = z$; for $z = x + iy$, $\tan z = (\sin 2x + i \sinh 2y)/(\cos 2x + \cosh 2y)$, and if $xy \neq 0$ the real and imaginary parts of $\tan z = z$ force $\sin 2x/(2x) = \sinh 2y/(2y)$, impossible since the left side is at most 1 in absolute value and the right exceeds 1 for $y \neq 0$; if $x = 0$, $\tanh y = y$ gives $y = 0$. So every nonzero root of $\tanh w = w$ is imaginary, and the first, $4.4934i$, lies outside $|w| < \pi$. So inside the disc there is exactly one critical point, it is the wall, and it is a saddle; the doubling interval, $w = \ln 2/2$, is an ordinary point of the same function with $f' = 0.112347\dots$. That is the same shape as §2's statement about the inflection – one, named, and not where the interval is. It is where the wall stands when both sides are in view.

4.4 · Why one function, in several places

§6 lists the fields in which p appears – a mean ratio, a midpoint rule, an acceptance probability, a tube coefficient, a Bernoulli generating function. The co-appearance has a reason, and it is short. The continuous homomorphisms $(\mathbb{R}, +) \rightarrow (\mathbb{R}_+, \times)$ are $x \mapsto e^{cx}$, and the normalisation $c = 1$ is the exponential used here; an interval measured by its length L and by the ratio of its ends $r = e^L$ is measured on the two sides of that map, and $p = GM/L_{\log}$ is the ratio of the two measures – the geometric mean is the multiplicative middle, the logarithmic mean is $(r - 1)/L$, the additive one. So wherever a problem carries both a count that adds and a step that compounds – panels and an exponential integrand; doublings and a probability; a translation length and a trace – $p(L)$ is present and is the same function. This places the recurrence; it does not place the ratio. Which fields carry both measures is a census, not a theorem.

Two checks that it is the object, and not a coordinate of it, that recurs. The relation $p(2L) = p(L)/\cosh(L/2)$: doubling the argument divides p by $\cosh(L/2)$. And the multiplier of the squaring map – $r \mapsto r^2$, which doubles the argument – at its fixed point, the wall: written $L \mapsto 2L$, $r \mapsto r^2$, $X \mapsto 2X/(1+X^2)$ or $\tau \mapsto 2\tau/(1+\tau^2)$ with $\tau = \tanh(L/2)$, the map's derivative at the wall is 2 in each – the fixed-point multiplier of the squaring map is invariant under smooth local conjugacy – where the second derivative of G at the same point was $1/3$, $1/12$, $1/12$, $1/3$ in those coordinates (§4.3). The multiplier is the object's; the curvature's value is the coordinate's.

§5 – THE ENCLOSURE

The bracket is not two numbers; it is the first order of a certified sequence. The certificate is three steps in integers feeding Leibniz.

Lemma. $|t_{k+1}|/|t_k| = (2k+1)^2/(16(k+1)(2k+3))$ for every $k \geq 1$. Proof. $C(2k+2, k+1)/C(2k, k) = (2k+2)(2k+1)/(k+1)^2 = 2(2k+1)/(k+1)$, and $|t_{k+1}|/|t_k| = [2(2k+1)/(k+1)] \cdot [(2k+1)/(32(2k+3))]$. ■

Two integer facts. $16(k+1)(2k+3) - (2k+1)^2 = 28k^2 + 76k + 47 > 0$, so the terms strictly decrease; $16(k+1)(2k+3) - 8(2k+1)^2 = 48k + 40 > 0$, so the ratio is below $1/8$.

Theorem. Since the terms alternate and strictly decrease, for every $j \geq 1$

$$S_{\{2j\}} < G < S_{\{2j-1\}} \quad \text{with width exactly } |t_{\{2j\}}|,$$

and strictly, because the tail after any partial sum lies strictly between 0 and the next term. (G is also transcendental, by Lindemann.)

$$\begin{array}{ll} j = 1 & 151/7680 < G < 1/48 \quad \text{width} \\ & 3/2560 \approx 1.17e-3 \\ j = 4 & 8311000980567041/420983760821944320 < G < \\ & 3819393966191/193466801848320 \quad \text{width} \\ & 6435/9345848836096 \approx 6.89e-10 \end{array}$$

Order 8 narrows order 2 by exactly $18253611008/10725 = 1701968.39\dots$. The asymptotic rate is $\log_{10} 8 = 0.903$ decimal digits per order, approached from above.

5.1 · The baseline this enclosure has to beat

By §1's midpoint identity, G is the relative error of the one-panel midpoint rule for e^t on $[0, \ln 2]$; absolute and relative coincide here only because $\int_0^{\ln 2} e^t dt = 1$ (at $r = 3$ they are 0.0971476982 and 0.0485738491). The textbook error form $E = (b-a)^3 f''(\xi)/24$ with $f'' \in (1, 2)$ gives, in one line,

$$(\ln 2)^3/24 = 0.0138760272 < G < 0.0277520543 = 2(\ln 2)^3/24$$

correct, two-sided, and reachable in thirty seconds. The first order of §5 is 11.84× narrower than that bracket, and every order after it gains 0.903 digits on the same certificate.

The Hermite–Hadamard literature has a bracket that does extend. Sándor (1988), as presented by Niculescu in Dragomir & Pearce (2000, Ch. 5, Corollary 60), refines $G \leq L_{\log} \leq A$ to $(a^{3/4}b^{1/4} + a^{1/4}b^{3/4})/2 < L_{\log} < (A + G)/2$, and the multiplicative mean-value lemma there ($M^*(f)^2 = M^*(f|[a, \sqrt{ab}]) \cdot M^*(f|[\sqrt{ab}, b])$, $f = \exp$) iterates it over the dyadic points $a^{1-k/2^n}b^{k/2^n}$. At $r = 2$ the first level gives $0.0148286 < G < 0.0294373$, and each level divides the width by about four: level 5 gives $0.0197371 < G < 0.0197514$, width 1.4×10^{-5} , from 32 subintervals (33 dyadic nodes). So the bracket on $L_{\log}$ and the lemma that iterates it are in print; the enclosure of this complement, $0.0148286 < G < 0.0294373$, is assembled from them here in one monotone step and appears on no page (entered from $L_{\log}$, a bound on a mean; entered from an interval fixed in advance, a bound on the object – the same distinction §4 draws for Bala). What §5 adds is rational bounds – integers over integers at every order – on an integer certificate with a closed-form term ratio, gaining 0.903 digits per term where the dyadic refinement gains 0.602 per level – and a term is one rational summand while a level doubles the number of points: at equal accuracy, eight terms against 8,192 subintervals.

$1/48$ and $151/7680$ are S_1 and S_2 , derived and certified here. No source we have found prints them as bounds on this object – a statement about our search, not about the literature.

§6 – WHERE IT SITS

Hermite (1881/83); Hadamard (1893)

where it is written: $f(\text{mid}) \leq \text{mean of } f \leq (f(\alpha) + f(\beta))/2$ for convex f ; at $f = \exp$ on the log interval, the chain $GM \leq L_{\log} \leq AM$
 what stands here: the left half, $p \leq 1$, is this note's starting point

Carlson (1966, 1972); Bhatia (2008); Jameson & Mercer (2019)
 where it is written: the same chain in mean-theoretic form; the hypergeometric mean family
 what stands here: at $L = \ln 2$ the ratio of the first two is $\sqrt{2} \cdot \ln 2$; the identification of the ratio with $L/(2 \sinh(L/2))$ is observed here

Lord (2014); Ricardo, Mahony, Burn, Miles, Subramaniam (Gazette Feedback, 2015)
 where it is written: $\ln 2 < 1/\sqrt{2}$ as a "non-calculator challenge", and eight elementary proofs in reply, four of the general form $\ln x < \sqrt{x} - 1/\sqrt{x}$; one via $\cosh(1/\sqrt{2}) > 5/4 = \cosh(\ln 2)$, one comparing series with weights $C(2n, n)/4^n$
 what stands here: $\sqrt{2} \cdot \ln 2 < 1$, i.e. $G > 0$, in elementary form – §2's positivity. No complement, series, or bound on $1 - \sqrt{2} \cdot \ln 2$ on those pages

Fishman (1996)
 where it is written: $p = \sqrt{2} \cdot \ln 2$, p. 188; $1 - p$ formed numerically, p. 189, no symbol
 what stands here: the value here, and the fact that its complement was reached before – the earliest we have found, stated as that

Marsaglia (1984), J. Amer. Statist. Assoc. 79(385), 218–221
 where it is written: the exact-approximation method: a sampling density decomposed as $p \cdot I_{[0,1]} + (1 - p) \cdot f^*$, $0 < p \leq \min f_X$ (as presented in Fishman §3.8, eq. 41); at the largest p the residual f^* stays nonnegative (eq. 52)
 what stands here: the unit in which $\sqrt{2} \cdot \ln 2$ is read in that literature – a probability, the weight of a uniform component. Algorithm EA is one instance of the method, and its p is this note's value

Ahrens & Dieter (1988); Hamilton (1998)
 where it is written: Algorithm EA: an exponential sample as $K \cdot \ln 2 + V$, V truncated exponential on $0 \leq v \leq \ln 2$ (Fishman eqs. 46–48); its saturation probability, sixteen digits
 what stands here: the same value (their digits are the nearest IEEE-754 double, verified, not stated by them); the doubling interval is their V-cell

Hodgson & Kerckhoff (2005)
 where it is written: the tube coefficient $\approx 1/0.980258$ in Thm 4.4
 what stands here: exactly $1/(\sqrt{2} \cdot \ln 2)$; their $1/(2\sqrt{2})$ is $\sinh(\frac{1}{2} \ln 2)$

Bala (2022), OEIS A002162
 where it is written: $\ln 2 = 2 \cdot \operatorname{arcsinh}(\sqrt{2}/4) = 2\sqrt{2} \cdot \sum \dots / 32^n$
 what stands here: the same series, read from the complement's end

Sándor (1988); Niculescu, in Dragomir & Pearce (2000), Ch. 5 Cor. 60
 where it is written: for $f = \exp$, the multiplicative Hermite–Hadamard inequality gives $G \leq L_{\log} \leq A$, refined to $(a^{\{3/4\}}b^{\{1/4\}} + a^{\{1/4\}}b^{\{3/4\}})/2 < L_{\log} < (A+G)/2$ and iterated over dyadic points
 what stands here: at $r = 2$: $0.0148286 < G < 0.0294373$, width $\div 4$ per level – their bracket on $L_{\log}$ and their lemma; the enclosure of the complement is assembled here from them. §5's differs in kind: rational, integer-certified, 0.903 digits per term

the midpoint rule (any text)
 where it is written: the one-panel rule and $E = (b-a)^3 f''(\xi)/24$
 what stands here: p is midpoint-over-exact for e^t on $[0, L]$; at $r = 2$ G is the rule's relative error, and $(\ln 2)^2/24$ is its first Bernoulli

term

the theory of $SL(2, \mathbb{Z})$; real quadratic units (any text)

where it is written: hyperbolic classes by trace; $r + 1/r \in \mathbb{Z}$ for the norm-one units of $\mathbb{Q}(\sqrt{n^2-4})$

what stands here: the integer points of t_2 in §4.2; the golden and silver points and the boundary are three of them

Chebyshev; the Lucas numbers (classical)

where it is written: $\cosh 2L = 2\cosh^2 L - 1$; L_{2^k} from 3

what stands here: the squaring of t_2 in §4.2: 3, 7, 47, 2207, ...; the Fermat ladder from 5/2

the metallic means (classical)

where it is written: $(m + \sqrt{m^2+4})/2$, norm -1

what stands here: their squares are the integer points $t_2 = m^2 + 2$ (§4.2)

the trapezoid rule (any text); Hermite–Hadamard's right member

where it is written: $T = (L/2) \cdot \coth(L/2) - 1$ for e^t

what stands here: the over-side of the same inequality whose under-side is G ; at the doubling interval $T = 3 \cdot \ln 2/2 - 1 = 0.0397207708\dots$, since $\coth(\ln 2/2) = 3$; $T/G \rightarrow 2$ at the wall (§7, second block)

harmonic functions (any text)

where it is written: a harmonic function has no interior extremum

what stands here: the saddle of $\operatorname{Re} f$ at the wall, §4.3

this note

where it is written: –

what stands here: the complement as an object: a symbol, its value, the central-binomial series read from this interval, the Bernoulli series in L , the term ratio in closed form, the integer certificate, the enclosure at every order

Each row separates what is theirs from what stands here; the last row is the claim. The negative in the last row is scoped: in Carlson (1966), Bhatia (2008) and Jameson & Mercer (2019), six probes each on 2026-08-27, no source gives the complement a symbol, a closed form, or bounds stated on $1 - p$. That search covered the means literature and not the Hermite–Hadamard / Jensen-gap literature. Its standing reference, Dragomir & Pearce, Selected Topics on Hermite–Hadamard Inequalities and Applications (RGMIA Monographs, 2000; 2002 edition), was read in full and searched on 2026-09-02 for the doubling ratio, the values $0.98025\dots$, $0.019741\dots$, $0.028481\dots$, $1.4427\dots$ beside $1.4142\dots$, $\sqrt{2} \cdot \ln 2$, and the phrase Jensen gap: zero occurrences. It contains the chain $H \leq G \leq L \leq I \leq A$ (its (2.6); there G and L are the geometric and logarithmic means, not this note's G and L), Pittenger's $A_{2/3} < I < A_{\ln 2}$ (where $\ln 2$ is a power-mean index, not this object), and Sándor's dyadic enclosure of $L_{\log}$, cited in §5.1. That literature writes the slack in absolute form, $1/\ln 2 - \sqrt{2}$; the search was for both numbers. A universal negative cannot be established by reading; one page reference closes the central claim.

§7 – REPRODUCTION

The complete runnable scripts accompany this paper as

The_complement_1_minus_sqrt2_ln2_block1.py and _block2.py; the listings are reproduced below. Run the files rather than a copy taken from the PDF – page extraction can drop a space of indentation, and the text file keeps it. mpmath is the only dependency; under a minute, most of it the dps 500 rung.

```

# -----
# 1 - sqrt(2)*ln(2) -- every claim in this paper, checked. G := 1 - sqrt(2)*ln(2)
# -----
from mpmath import mp, mpf, sqrt, log, exp, sinh, cosh, asinh, diff, findroot, pi, quad
from fractions import Fraction as Q
from math import comb, comb as fcomb, factorial as ffact

CH = FA = 0
def ck(name, cond):
    global CH, FA
    CH += 1
    if not cond: FA += 1
    print(f" [{ 'ok' if cond else 'FAIL' }] {name}")

def term(k):
    # k >= 1, exact rational
    return Q((-1)**(k+1) * comb(2*k, k), 32**k * (2*k+1))

for DPS in (30, 60, 200, 500):
    mp.dps = DPS
    print(f"\n--- dps {DPS} " + "-"*52)
    L = log(mpf(2)) # the interval: r = 2
    X = (mpf(2) - 1) / (2*sqrt(mpf(2))) # half-difference coordinate
    p_ln2 = sqrt(mpf(2)) * L # the ratio p at r = 2
    G = 1 - p_ln2
    eps = mpf(10)**(-(DPS - 8))

    # 1. the interval
    ck("sinh(ln2 / 2) = 1/(2 sqrt2)", abs(X - sinh(L/2)) < eps)
    ck("X^2 = 1/8 exactly", abs(X**2 - mpf(1)/8) < eps)
    ck("arsinh(1/(2 sqrt2)) = ln2 / 2", abs(asinh(X) - L/2) < eps)
    ck("p(ln2) = arsinh(X)/X = sqrt2 * ln2", abs(asinh(X)/X - p_ln2) < eps)
    ck("GM/L_log on [1,2] equals L/(2 sinh(L/2))",
        abs(sqrt(mpf(2))/((mpf(2)-1)/L) - L/(2*sinh(L/2))) < eps)

    # 1b. the chain IS Hermite-Hadamard for exp on the log interval (section 1)
    def HH(a, b):
        # the three H-H terms at f = exp
        la, lb = log(mpf(a)), log(mpf(b))
        return (exp((la+lb)/2),
                quad(exp, [la, lb])/(lb-la),
                (exp(la)+exp(lb))/2)
        # f((alpha+beta)/2)
        # mean of f
        # (f(alpha)+f(beta))/2
    def means(a, b):
        a, b = mpf(a), mpf(b)
        return sqrt(a*b), (b-a)/log(b/a), (a+b)/2 # GM, L_log, AM
    ck("Hermite-Hadamard at f=exp on [ln a, ln b] gives GM, L_log, AM exactly",
        all(max(abs(h-m) for h, m in zip(HH(a,b), means(a,b))) < eps
            for a, b in ((1,2), (1,3), (2,7), ('0.5','11.3'))))
    ck("so p = GM/L_log is H-H's LEFT half, and p <= 1 is that inequality",
        all(mpf(0) < means(a,b)[0]/means(a,b)[1] <= 1 for a, b in ((1,2), (1,3), (2,7))))
    ck("ABSOLUTE Jensen gap at r = 2 = 1/ln2 - sqrt2 = 0.02848147851586835855",
        abs((mpf(2)-1)/L - sqrt(mpf(2)) - mpf('0.02848147851586835855')) < mpf('1e-20'))
    ck("and G is that gap NORMALISED: (L_log - GM)/L_log = (1/ln2 - sqrt2)*ln2",
        abs(((mpf(2)-1)/L - sqrt(mpf(2))) / ((mpf(2)-1)/L) - G) < eps
        and abs((1/L - sqrt(mpf(2)))*L - G) < eps)
    ck("the two are DIFFERENT numbers -- the qualifier 'relative' is load-bearing",
        abs(((mpf(2)-1)/L - sqrt(mpf(2))) - G) > mpf('0.008'))
    ck("mean-ratio and midpoint are ONE setting: GM/L_log = midpt/exact, at every L",
        all(abs(means(1, exp(mpf(t)))[0]/means(1, exp(mpf(t)))[1]
            - mpf(t)*exp(mpf(t)/2)/(exp(mpf(t))-1)) < eps
            for t in ('0.6931471805599453', '1.5', '4'))))

```

```

# 2. positivity, before any evaluation
ck("G > 0 (arsinh strictly concave, arsinh(0)=0)", G > 0)

# 3. the series
S = sum((term(k) for k in range(1, 4*DPS)), Q(0))
ck("series -> G", abs(mpf(S.numerator)/S.denominator - G) < eps)
ck("S1 = 1/48", term(1) == Q(1, 48))
ck("S1 + S2 = 151/7680 (exact integers)", term(1)+term(2) == Q(151, 7680))
ck("bracket width = 3/2560 = 0.001171875", term(1)-(term(1)+term(2)) == Q(3, 2560))

# 4. the ratio lemma, and the two integer facts under it
ck("central-binomial step C(2k+2,k+1)/C(2k,k) = 2(2k+1)/(k+1) [proved in 5]",
    all(Q(comb(2*k+2, k+1), comb(2*k, k)) == Q(2*(2*k+1), k+1) for k in range(1, 41)))
ck("|t(k+1)|/|t(k)| = (2k+1)^2 / (16(k+1)(2k+3)), k=1..40",
    all(abs(term(k+1))/abs(term(k)) == Q((2*k+1)**2, 16*(k+1)*(2*k+3)) for k in range(1,
    41)))
ck("16(k+1)(2k+3) - (2k+1)^2 = 28k^2+76k+47 > 0 -> decreasing",
    all(16*(k+1)*(2*k+3) - (2*k+1)**2 == 28*k**2+76*k+47 > 0 for k in range(0, 200)))
ck("16(k+1)(2k+3) - 8(2k+1)^2 = 48k+40 > 0 -> ratio < 1/8",
    all(16*(k+1)*(2*k+3) - 8*(2*k+1)**2 == 48*k+40 > 0 for k in range(0, 200)))

# 5. the enclosure
P = [sum((term(i) for i in range(1, j+1)), Q(0)) for j in range(1, 13)]
q = lambda f: mpf(f.numerator)/f.denominator
ck("S(2j) < G < S(2j-1) for j = 1..6",
    all(q(P[2*j-1]) < G < q(P[2*j-2]) for j in range(1, 7)))
ck("bracket width at step j is exactly |t(2j)|, j = 1..6",
    all(P[2*j-2]-P[2*j-1] == abs(term(2*j)) for j in range(1, 7)))
ck("S7/S8 narrow the first bracket by exactly 18253611008/10725",
    (Q(3,2560) / (P[6]-P[7]) == Q(18253611008, 10725)))
ck("the j=4 bracket AS PRINTED in section 5, numerators and denominators",
    P[7] == Q(8311000980567041, 420983760821944320)
    and P[6] == Q(3819393966191, 193466801848320)
    and P[6]-P[7] == Q(6435, 9345848836096))

# 6. the function does not select the interval (section 2)
# the sign facts below are PROVED in section 2; these are regression tests.
Gf = lambda t: 1 - t/(2*sinh(t/2))
Gp = lambda t: (t/2*cosh(t/2) - sinh(t/2)) / (2*sinh(t/2)**2)
psi = lambda t: 4*sinh(t) - t*cosh(t) - 3*t
pts = ('0.01','0.5','0.6931471805599453','3','10','29')
ck("G'(L) = (u cosh u - sinh u)/(2 sinh^2 u), u = L/2",
    all(abs(diff(Gf, mpf(t)) - Gp(mpf(t))) < eps for t in pts))
ck("its numerator f(u) = u cosh u - sinh u > 0 [f(0)=0, f' = u sinh u > 0]",
    all(mpf(t)/2*cosh(mpf(t)/2) - sinh(mpf(t)/2) > 0 for t in pts))
ck("so G'(L) > 0 -- no extremum anywhere on (0, inf)",
    all(Gp(mpf(t)) > 0 for t in pts))
ck("G''(L) = psi(L)/(16 sinh^3(L/2)), psi(L) = 4 sinh L - L cosh L - 3L",
    all(abs(diff(Gf, mpf(t), 2) - psi(mpf(t))/(16*sinh(mpf(t)/2)**3)) < eps for t in pts))
if DPS >= 30:
    Ls = findroot(psi, mpf('3.2'))
    ck("the inflection is psi's unique positive root, L* = 3.2122305976",
        abs(Ls - mpf('3.212230597605534728')) < mpf('1e-15'))
    ck("and it is a zero of G'' itself (r = e^L* = 24.834)",
        abs(diff(Gf, Ls, 2)) < eps and abs(exp(Ls) - mpf('24.8344200858')) < mpf('1e-9'))
    ck("psi's uniqueness cascade: v1 < v2 < v3 < L*",
        (lambda v1, v2, v3: v1 < v2 < v3 < Ls)(
            findroot(lambda v: cosh(v) - v*sinh(v), mpf('1.2')),
            findroot(lambda v: diff(psi, v, 2), mpf('1.9')),
            findroot(lambda v: diff(psi, v), mpf('2.6'))))
    ck("L/2pi = 0.110317800076 IS the decay factor, not a distance:"
        " 7*(L/4pi)^2 = (7/4)*(L/2pi)^2",

```

```

    abs(L/(2*pi) - mpf('0.110317800076')) < mpf('1e-12')
    and abs(7*(L/(4*pi))**2 - (mpf(7)/4)*(L/(2*pi))**2) < eps)
# the inflection is NOT coordinate-free -- section 2 states both and quotes
# no distance to either. Gr = G as a function of r.
Gr = lambda t: 1 - log(t)/(2*sinh(log(t)/2))
rs = findroot(lambda t: diff(Gr, t, 2), mpf('2.4'))
ck("in r A root sits at 2.474062197, NOT e^L* = 24.834 -- uniqueness in r NOT proved",
    abs(rs - mpf('2.474062197326598')) < mpf('1e-12')
    and abs(rs - exp(Ls)) > 20)
ck("no extremum in r either (dG/dr > 0; L = ln r is strictly increasing)",
    all(diff(Gr, mpf(t)) > 0 for t in ('1.05', '2', '5', '24.834', '1000'))))

# 7. what other people printed
ck("Fishman 1996 p.189: 100 * 1.511076 * G = 2.983144 (as printed)",
    abs(100*mpf('1.511076')*G - mpf('2.983144')) < mpf('1e-6'))
ck("section 3: that product is 2.98314456..., so his 2.983144 is TRUNCATED (rounded "
    "gives ...45)",

    abs(100*mpf('1.511076')*G - mpf('2.98314456')) < mpf('5e-9')
    and mp.nstr(100*mpf('1.511076')*G, 7) == '2.983145')
ck("section 3: the alpha reproducing 2.983144 exactly is 1.5110757...",
    abs(mpf('2.983144')/(100*G) - mpf('1.5110757')) < mpf('5e-8'))
ck("section 2: at the r-inflection 2.474062197 the argument is L = 0.905861414",
    abs(log(mpf('2.474062197326598')) - mpf('0.905861414')) < mpf('5e-10'))
G3 = 1 - log(mpf(3))/(2*sinh(log(mpf(3))/2)) # G(3); the integral on [0, ln 3] is 2
ck("section 5.1: at r = 3 the absolute and relative midpoint errors are 0.0971476982 "
    "and 0.0485738491",

    abs(2*G3 - mpf('0.0971476982')) < mpf('5e-11')
    and abs(G3 - mpf('0.0485738491')) < mpf('5e-11'))
ck("Hodgson-Kerckhoff's coefficient X/arcsinh(X) = 1/(sqrt(2) ln 2)",
    abs(X/asinh(X) - 1/p_ln2) < eps)
ck("Bala A002162 clause 1: 2*arcsinh(sqrt(2)/4) = ln 2",
    abs(2*asinh(sqrt(mpf(2))/4) - L) < eps)
ck("Bala's argument sqrt(2)/4 is this interval's X",
    abs(sqrt(mpf(2))/4 - X) < eps)
ck("chart domains: X = 1 exactly at r = 3 + 2*sqrt(2) = (1+sqrt(2))^2",
    abs((3+2*sqrt(mpf(2))-1)/(2*sqrt(3+2*sqrt(mpf(2)))) - 1) < eps)

# --- 4, the base IS the coordinate -----
Xof = lambda r: (r - 1)/(2*sqrt(r))
baseof = lambda r: 16*r/(r - 1)**2
ck("base = 4/X^2 exactly, at r = 2, 3, 5, 3+2sqrt(2)",
    all(abs(baseof(r) - 4/Xof(r)**2) < eps
        for r in (mpf(2), mpf(3), mpf(5), 3+2*sqrt(mpf(2)))))
ck("integer ratios with integer base: n = 2, 3, 5 (bases 32, 12, 5); power of two only "
    "at n = 2",

    [n for n in range(2, 40) if 16 % (n-1)**2 == 0] == [2, 3, 5]
    and [int(baseof(mpf(n))) for n in (2, 3, 5)] == [32, 12, 5])
ck("the domain bound r <= 3+2sqrt(2) IS base = 4, not a separate fact",
    abs(baseof(3+2*sqrt(mpf(2)))) - 4) < eps)
# (checks on other ratios -- r = phi^2, 2+sqrt(3), the shared numerators -- belong to the family
# and are not printed in this note.)
ck("Bernoulli radius |L/2| < pi <=> r < e^(2pi) = 535.4916..., and 2 is inside",
    abs(exp(2*pi) - mpf('535.4916552476473')) < mpf('1e-10')
    and log(2) < 2*pi)
ck("midpoint/exact = L*e^(L/2)/(e^L - 1) = p(L) at r = 2, 3, 5, 10",
    all(abs(log(r)*exp(log(r)/2)/(exp(log(r))-1)
        - log(r)/(2*sinh(log(r)/2))) < eps for r in (2, 3, 5, 10)))
Gm2, Am2 = sqrt(mpf(2)), mpf(3)/2 # Sandor / Niculescu, D&P Cor. 60, at r = 2
lo2 = (mpf(2)**(mpf(1)/4) + mpf(2)**(mpf(3)/4))/2

```

```

hi2 = (Am2 + Gm2)/2
ck("section 5.1: Sandor's refinement at r=2 brackets G: 0.0148286 < G < 0.0294373",
    lo2 < 1/log(2) < hi2 and abs((1 - Gm2/lo2) - mpf('0.0148286')) < mpf('5e-8')
    and abs((1 - Gm2/hi2) - mpf('0.0294373')) < mpf('5e-8')
    and (1 - Gm2/lo2) < G < (1 - Gm2/hi2))
ck("(ln2)^2/24 carries 101.403% of G",
    abs((L**2/24)/G - mpf('1.01403')) < mpf('1e-5'))
ck("section 4.1: (ln2)^2/24 = 0.0200188756, as printed",
    abs(L**2/24 - mpf('0.0200188756')) < mpf('5e-11'))
ck("section 5: the narrowing factor 18253611008/10725 = 1701968.39..., as printed",
    abs(mpf(18253611008)/10725 - mpf('1701968.39')) < mpf('5e-3'))

# --- 4.1 the Bernoulli route -----
def bern(m):
    # B_j exact, from the recurrence
    B = [Q(0)] * (m + 1); B[0] = Q(1)
    for n in range(1, m + 1):
        B[n] = -sum(Q(fcomb(n + 1, j)) * B[j] for j in range(n)) / Q(n + 1)
    return B
# 24: the 8-term sum needs B_16, the sporadic-agreement check needs B_24
B = bern(24)
ck("Bernoulli recurrence gives B2,B4,B6,B8 = 1/6, -1/30, 1/42, -1/30",
    (B[2], B[4], B[6], B[8]) == (Q(1,6), Q(-1,30), Q(1,42), Q(-1,30)))

def bcoef(n):
    # coefficient of L^(2n) in G(L)
    return Q(2**(2*n) - 2) * B[2*n] / Q(2**(2*n) * ffact(2*n))
ck("G(L) coefficients are 1/24, -7/5760, 31/967680, -127/154828800",
    [bcoef(n) for n in (1,2,3,4)]
    == [Q(1,24), Q(-7,5760), Q(31,967680), Q(-127,154828800)])
ck("section 4.1: num(c_6) = 1414477 as printed; num(c_9) = 5749691557 beyond it",
    abs(bcoef(6).numerator) == 1414477 and abs(bcoef(9).numerator) == 5749691557)

def bmp(n):
    c = bcoef(n); return mpf(c.numerator)/mpf(c.denominator)
ck("the Bernoulli series reproduces G from B_2n alone (8 terms)",
    abs(sum(bmp(n) * L**(2*n) for n in range(1, 9)) - G) < mpf('1e-16'))

ck("c_n carries the factor 2^(2n-1)-1 ALWAYS; only the cancellation varies",
    all(bcoef(n) == Q(2**(2*n-1)-1) * B[2*n] / Q(2**(2*n-1) * ffact(2*n))
        for n in range(1, 11)))
ck("num(c_n) = M * num(B_2n) / (cancellation vs (2n)!) -- TWO mechanisms",
    all((2**(2*n-1)-1) * abs(B[2*n].numerator) % abs(bcoef(n).numerator) == 0
        and ffact(2*n) % ((2**(2*n-1)-1) * abs(B[2*n].numerator)
            // abs(bcoef(n).numerator)) == 0
        for n in range(1, 13)))
ck("agreement num(c_n)=M is SPORADIC: n = 1,2,3,4 and 7, not a threshold",
    [n for n in range(1, 13) if abs(bcoef(n).numerator) == 2**(2*n-1)-1]
    == [1, 2, 3, 4, 7])
ck("at n=6 NOTHING cancels and the numerator is still not M: 2047*691",
    abs(bcoef(6).numerator) == 2047*691 and 2047*691 != 2047)
ck("prefactor identity (2^(2n+1)-1)/(2^(2n-1)-1) = 4 + 3/(2^(2n-1)-1)",
    all(Q(2**(2*n+1)-1, 2**(2*n-1)-1) == 4 + Q(3, 2**(2*n-1)-1) for n in range(1, 41)))
ck("so it falls strictly from 7 at n=1 toward 4 -- max is 7, by inspection",
    Q(2**3 - 1, 2**1 - 1) == 7
    and all(Q(2**(2*n+3)-1, 2**(2*n+1)-1) < Q(2**(2*n+1)-1, 2**(2*n-1)-1) > 4
        for n in range(1, 41)))
ck("ratio bound 7*(ln2/(4*pi))^2 = 0.0212975297739, and is < 1",
    abs(7*(L/(4*pi))**2 - mpf('0.0212975297739')) < mpf('1e-12')
    and 7*(L/(4*pi))**2 < 1)

A = [sum(bmp(n) * L**(2*n) for n in range(1, j+1)) for j in range(1, 9)]
Sb = [sum(mpf(term(k).numerator)/mpf(term(k).denominator)
    for k in range(1, j+1)) for j in range(1, 9)]

```

```

ck("the Bernoulli bracket lies strictly inside the S_k bracket, orders 2-8",
    all(min(Sb[j-1], Sb[j-2]) < min(A[j-1], A[j-2])
        and max(A[j-1], A[j-2]) < max(Sb[j-1], Sb[j-2]) for j in range(2, 9)))
ck("and the width factor at 8 terms (§5's j=4) exceeds 7e5",
    abs(Sb[7]-Sb[6]) / abs(A[7]-A[6]) > mpf('7e5'))

print("\n" + "="*64)
print(f"  CHECKS {CH}    FAILURES {FA}")
print("="*64)

```

```

=====
CHECKS 260    FAILURES 0
=====

```

Second block – the where (§4.2–4.4, Figure 1). Standalone; the file _block2.py.

```

# -----
# THE WHERE -- every number in §4.2–§4.4 and Figure 1, checked.    (second block; standalone)
# -----
from mpmath import (mp, mpf, mpc, sqrt, log, exp, sinh, cosh, tanh, asinh, atan, acos, coth, pi,
    diff, re, im, nstr)
CH2 = FA2 = 0
def ck2(name, cond):
    global CH2, FA2
    CH2 += 1
    if not cond: FA2 += 1; print("  FAIL:", name)
for dps in (30, 60, 200, 500):
    mp.dps = dps; eps = mpf(10)**(-(dps-8))
    p = lambda L: L/(2*sinh(L/2))
    G = lambda r: 1 - p(log(r))
    X = lambda r: (r-1)/(2*sqrt(r))
    t2 = lambda r: r + 1/r                # the square's trace, 2 cosh L
    t = lambda r: sqrt(r) + 1/sqrt(r)      # the trace, 2 cosh(L/2)
    th = lambda r: atan(sinh(log(r)))      # theta = gd(L)
    partner = lambda r: exp(asinh(1/sinh(log(r))))
    Pi = lambda r: acos(tanh(log(r)/2))    # angle of parallelism (half argument)
    deg = lambda a: a*180/pi
    phi = (1+sqrt(5))/2; dS = 1+sqrt(2)
    # --- §4.2 the line and the cells
    ck2("t2 = t^2 - 2 (Cayley-Hamilton) at r = 2, phi^2, 3+2sqrt2",
        all(abs(t2(r)-(t(r)**2-2))<eps for r in (mpf(2), phi**2, dS**2)))
    ck2("doubling: t2 = 5/2, between the wall (2) and golden (3); X^2 = 1/8",
        abs(t2(mpf(2))-mpf(5)/2)<eps and abs(X(mpf(2))**2-mpf(1)/8)<eps)
    ck2("golden r = phi^2: t2 = 3, chord 2X = 1, p = L exactly, G = 1 - 2 ln phi",
        abs(t2(phi**2)-3)<eps and abs(2*X(phi**2)-1)<eps
        and abs(p(log(phi**2))-log(phi**2))<eps
        and abs(G(phi**2)-(1-2*log(phi)))<eps)
    ck2("2+sqrt3: t2 = 4, base 8, theta = 60 deg", abs(t2(2+sqrt(3))-4)<eps
        and abs(4/X(2+sqrt(3))**2-8)<eps and abs(th(2+sqrt(3))-pi/3)<eps)
    ck2("the boundary 3+2sqrt2 = (1+sqrt2)^2: t2 = 6, X = 1, base 4, theta = arccos(1/3)",
        abs(dS**2-(3+2*sqrt(2)))<eps and abs(t2(dS**2)-6)<eps and abs(X(dS**2)-1)<eps
        and abs(th(dS**2)-acos(mpf(1)/3))<eps)
    ck2("metallic means squared are the cells t2 = m^2 + 2, m = 1..5",
        all(abs(t2(((m+sqrt(m*m+4))/2)**2)-(m*m+2))<eps for m in range(1,6)))
    ck2("the base is 16/(t2 - 2): 32, 16, 12, 8, 5, 4 at r = 2, phi^2, 3, 2+sqrt3, 5, 3+2sqrt2",
        all(abs(4/X(r)**2-16/(t2(r)-2))<eps for r in (mpf(2), phi**2, mpf(3), 2+sqrt(3), mpf(5),
            dS**2)) and
        [round(float(4/X(r)**2)) for r in (mpf(2), phi**2, mpf(3), 2+sqrt(3), mpf(5), dS**2)] == [32,
            16, 12, 8, 5, 4])
L = [mpf(3)]

```

```

for _ in range(4): L.append(L[-1]**2-2)
ck2("the square's trace under squaring, from golden: 3, 7, 47, 2207, 4870847 (Lucas "
    "L_{2^k})", [int(x) for x in L]==[3,7,47,2207,
    4870847])
F = [mpf(5)/2]
for _ in range(3): F.append(F[-1]**2-2)
ck2("from doubling: 5/2, 17/4, 257/16, 65537/256 -- Fermat numbers over 2^(2^k); never "
    "an integer",
    all(abs(F[k]-mpf(2**(2**(k+1))+1)/2**(2**k))<eps for k in range(4))
    and all(abs(x-round(x))>mpf('1e-3') for x in F))
ck2("the pairing sinh L * sinh L' = 1: 2 <-> 3, phi^2 <-> sqrt5, 3+2sqrt2 <-> sqrt2, "
    "1+sqrt2 <-> itself",
    abs(partner(mpf(2))-3)<eps and abs(partner(phi**2)-sqrt(5))<eps
    and abs(partner(dS**2)-sqrt(2))<eps and abs(partner(dS)-dS)<eps)
ck2("gd(ln 2) + gd(ln 3) = 90 deg: the 3-4-5 triangle and its complement",
    abs(th(mpf(2))+th(mpf(3))-pi/2)<eps
    and abs(cosh(0)*mp.cos(th(mpf(2)))-mpf(4)/5)<eps)
# --- §4.2 the angles (added 2026-09-15)
# (r, t2 as printed, G as printed, theta as printed, Pi as printed) -- the table's rows
cells = [(mpf(1), mpf(2), "0", "0", "90"),
    (mpf(2), mpf(5)/2, "0.019741857", "36.86989765", "70.52877937"),
    (phi**2, mpf(3), "0.037576350", "48.18968510", "63.43494882"),
    (mpf(3), mpf(10)/3, "0.048573849", "53.13010235", "60"),
    (2+sqrt(3), mpf(4), "0.068770141", "60", "54.73561032"),
    ((5+sqrt(21))/2, mpf(5), "0.095408039", "66.42182152", "49.10660535"),
    (mpf(5), mpf(26)/5, "0.100296856", "67.38013505", "48.18968510"),
    (dS**2, mpf(6), "0.118626413", "70.52877937", "45")]
ck2("cos theta = 2/t2 and cos Pi = (r-1)/(r+1) at the eight cells of the table; every "
    "printed entry of the table (t2, G, theta, Pi) reproduced from the closed form",
    all(abs(1/cosh(log(r))-2/t2(r))<eps and abs(tanh(log(r)/2)-(r-1)/(r+1))<eps
    and abs(t2(r)-tt)<eps
    and abs((mpf(0) if r == 1 else G(r))-mpf(g))<mpf("5e-10") # nine places
    and abs(deg(th(r))-mpf(a))<mpf("5e-9") # eight places, as printed
    and abs(deg(Pi(r))-mpf(b))<mpf("5e-9")
    for r,tt,g,a,b in cells))
ck2("golden: theta = arccos(2/3), Pi = arctan 2; Pi(2) = theta(3+2sqrt2) = arccos(1/3), "
    "Pi(3) = theta(2+sqrt3) = 60 deg, Pi(5) = theta(phi^2): t2(r') = 2 + 4/(r-1), an "
    "integer iff (r-1) | 4",
    abs(th(phi**2)-acos(mpf(2)/3))<eps and abs(Pi(phi**2)-atan(mpf(2)))<eps
    and abs(Pi(mpf(2))-th(dS**2))<eps and abs(Pi(mpf(3))-th(2+sqrt(3)))<eps
    and abs(Pi(mpf(3))-pi/3)<eps and abs(Pi(mpf(5))-th(phi**2))<eps
    and all(abs(t2(rp)-(2+mpf(4)/(r-1)))<eps
    for r,rp in ((2,dS**2),(3,2+sqrt(3)),(5,phi**2)))
    and [r for r in range(2,61) if 4 % (r-1) == 0] == [2,3,5])
# --- §4.3 the saddle
Gv = lambda v: mpf(0) if v == 0 else 1 - v/sinh(v) # v = L/2
h = mpf(10)**(-(dps//4))
d2 = lambda f, x0: (f(x0+h)-2*f(x0)+f(x0-h))/h**2
ck2("G'' at the wall: 1/3 in v = L/2, 1/12 in L (the value is a coordinate)", abs(d2(Gv,
    mpf(0))-mpf(1)/3)<mpf('1e-6') and abs(d2(lambda L: Gv(L/2),
    mpf(0))-mpf(1)/12)<mpf('1e-6'))
f = lambda w: mpf(0) if w == 0 else 1 - w/sinh(w)
Re = lambda a,b: re(f(mpc(a,b)))
Faa = (Re(h,0)-2*Re(0,0)+Re(-h,0))/h**2; Fbb = (Re(0,h)-2*Re(0,0)+Re(0,
    -h))/h**2; Fab = (Re(h,h)-Re(h,-h)-Re(-h,h)+Re(-h,-h))/(4*h*h)
ck2("Re f at w = 0: Hessian diag(+1/3, -1/3), a saddle; the imaginary axis is the "
    "circular side",
    abs(Faa-mpf(1)/3)<mpf('1e-6'))

```

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and abs(Fbb+mpf(1)/3)<mpf('1e-6')
and abs(Fab)<mpf('1e-6'))
ck2("on the imaginary axis  $f(i \beta/2) = 1 - \beta/(2 \sin(\beta/2))$ : the circle's "
"argument over chord, negative", abs(re(f(mpc(0,
pi/4)))-(1-(pi/2)/(2*mp.sin(pi/4))))<eps and re(f(mpc(0,
pi/4)))<0)
lap = lambda a,b: (Re(a+h,b)+Re(a-h,b)+Re(a,b+h)+Re(a,b-h)-4*Re(a,b))/h**2
ck2("Re f is harmonic (Laplacian 0) at three interior points", all(abs(lap(a,
b))<mpf('1e-5') for a,b in ((mpf('0.3'),mpf('0.2')), (mpf('0.7'),
mpf('-0.4')), (mpf('1.1'),
mpf('0.9')))))
fp = lambda w: (w*cosh(w)-sinh(w))/sinh(w)**2
ck2("the doubling interval is an ordinary point:  $f'(\ln 2/2) = 0.112347305659$ ;  $f'(w) = 0$  "
"iff  $\tanh w = w$ ",
abs(fp(log(2)/2)-mpf('0.112347305659'))<mpf('1e-11'))
ys = [mp.findroot(lambda y: mp.tan(y)-y, x0) for x0 in (4.5, 7.7)]
ck2("the next critical points are  $w = \pm 4.4934i, \pm 7.7253i$  ( $\tan y = y$ ), beyond the pole "
"at  $i\pi$ : the wall is the only one in  $|w| < \pi$ ",
abs(ys[0]-mpf('4.493409458'))<mpf('1e-8') and abs(ys[1]-mpf('7.725251837'))<mpf('1e-8')
and ys[0] > pi)
Ab = lambda a,b: abs(f(mpc(a,b)))
ck2("|f| has a minimum at the wall (both second derivatives  $+1/3$ )", abs((Ab(h,0)-2*Ab(0,
0)+Ab(-h,0))/h**2-mpf(1)/3)<mpf('1e-5') and abs((Ab(0,h)-2*Ab(0,0)+Ab(0,
-h))/h**2-mpf(1)/3)<mpf('1e-5'))
# --- §4.4 why: the squaring map's multiplier is invariant, the Hessian's value is not
ck2("the squaring map  $r \rightarrow r^2$  ( $L \rightarrow 2L$ ): multiplier at its fixed point, the wall, is 2 "
"in L, r, X and tau", all(abs(diff(g,x0)-2)<mpf('1e-12') for g,x0 in ((lambda L:2*L,
0),(lambda r:r*r,1),(lambda x:2*x*sqrt(1+x*x),0),(lambda s:2*s/(1+s*s),
0))))
ck2("section 4.3:  $1 - v/\sinh v = v^2/6 - 7v^4/360 + 31v^6/15120$  (the series of 4.1 in v "
"=  $L/2$ )",
(lambda v: abs((1-v/sinh(v))-(v**2/6-7*v**4/360+31*v**6/15120))<v**8)(mpf("0.01")))
ck2("p(2L) = p(L)/cosh(L/2): doubling the argument divides p by cosh(L/2)",
all(abs(p(2*L_)-p(L_)/cosh(L/2))<eps for L_ in (log(2), log(3), mpf('0.1'))))
ck2("the trapezoid twin:  $T = (L/2)\coth(L/2) - 1 = 3 \ln 2/2 - 1 = 0.0397$  at doubling "
"(coth(ln2/2) = 3);  $T/G \rightarrow 2$  at the wall",
abs(((log(2)/2)*coth(log(2)/2)-1)-mpf('0.039720771'))<mpf('1e-9')
and abs(((log(2)/2)*coth(log(2)/2)-1)-(3*log(2)/2-1))<eps
and abs(coth(log(2)/2)-3)<eps
and abs(((mpf('1e-3')/2)*coth(mpf('1e-3')/2)-1)/Gv(mp('5e-4'))-2)<mpf('1e-5'))
# --- Figure 1
ck2("Figure 1's four points:  $G = 0, 0.019741857, 0.037576350, 0.118626413$ ; bases inf, "
"32, 16, 4",
abs(G(phi**2)-mpf('0.037576350'))<mpf('1e-9')
and abs(G(dS**2)-mpf('0.118626413'))<mpf('1e-9')
and abs(G(mpf(2))-mpf('0.019741857'))<mpf('1e-9'))
print(f"\n THE WHERE: CHECKS {CH2} FAILURES {FA2}")

```

THE WHERE: CHECKS 100 FAILURES 0

Accounting: the first block defines 65 distinct checks and the second 25, each run at dps 30, 60, 200 and 500: $90 \times 4 = 360$ executions. Checks on other ratios ($r = \phi^2, 2 + \sqrt{3}$, the shared numerators) belong to the family and are not printed here; the check that the integer ratios with integer base are $n = 2, 3, 5$, and Sándor's bracket at $r = 2$, are included because §4 and §5.1 print

them.

WHAT IS KNOWN

Every line is a calculation, a proof, or a citation. §7 regression-checks the printed numerical and algebraic identities (90 checks, 360 executions, zero failures); the universal statements rest on the proofs in the text, and the citations were checked against the pages. Each mathematical statement is exact, except one, labelled.

THE CHAIN IS HERMITE–HADAMARD AT $f = \exp$	an identification; places the object.
$G(r) > 0$ FOR EVERY $r > 1$	$\operatorname{arsinh} x < x$, by its integral; elementary and known as such (§2, §6).
NO EXTREMUM, ONE INFLECTION IN L	$G' > 0$; ψ 's cascade; $L^* = 3.2122305976$. classical, here because §2 needs it.
THE SERIES IS EXACT AND RATIONAL	$X^2 = 1/8$, so no irrational survives a term.
THE TERM RATIO IN CLOSED FORM	$(2k+1)^2/(16(k+1)(2k+3))$.
THE CERTIFICATE	$28k^2 + 76k + 47 > 0$ and $48k + 40 > 0$, by inspection.
THE ENCLOSURE	$S_{\{2j\}} < G < S_{\{2j-1\}}$ at every order, width $ t_{\{2j\}} $, strict.
A SECOND ENCLOSURE	the Bernoulli series in L , ratio under $7(\ln 2/4\pi)^2$.
THE TWO ARE NESTED – OBSERVATION	at orders 2–8, tested; not proved. the one line that is not a theorem.
THE MIDPOINT IDENTITY	G is the one-panel midpoint rule's relative error for e^t on $[0, \ln 2]$, exactly.
ONE EXACT EVALUATION ELSEWHERE	Hodgson & Kerckhoff's coefficient is $1/(\sqrt{2} \cdot \ln 2)$.
THE CONCORDANCE	§6: where each piece is written, and what stands here in each case. searched and dated; the negatives scoped to the pages read.
THE TWO TRACES	$t = 2\cosh(L/2)$, $t_2 = r + 1/r = t^2 - 2$; base = $16/(t_2 - 2)$. classical.
WHERE $r = 2$ STANDS	$t_2 = 5/2$, between the wall and the golden point; closes on the chord coordinate ($X^2 = 1/8$, base 32), not on the trace. computed.
THE PARTNER	$\sinh L \cdot \sinh L' = 1$ pairs doubling with tripling; $\operatorname{gd}(\ln 2) + \operatorname{gd}(\ln 3) = 90^\circ$. computed.
THE ANGLES	$\cos \theta = 2/t_2$ ($\theta = \operatorname{gd} L$) and $\cos \Pi = (r - 1)/(r + 1)$; Π at $r = 2, 3, 5$ is θ at the boundary, $2+\sqrt{3}$, ϕ^2 , under $(r - 1) \mid 4$. computed.
THE SADDLE	$\operatorname{Re}(1 - w/\sinh w)$ has Hessian $\operatorname{diag}(1/3, -1/3)$ at $w = 0$; harmonic, hence a saddle; the

THE HESSIAN'S VALUE IS A COORDINATE

WHY p RECURS

only critical point in $|w| < \pi$. classical + proved.
1/3 in $L/2$, 1/12 in L ; the sign is invariant, the number is not. classical.
the continuous homomorphisms $(\mathbb{R}, +) \rightarrow (\mathbb{R}_+, \times)$ are e^{cx} ; p is the ratio of the two measures. a placement of the recurrence.
